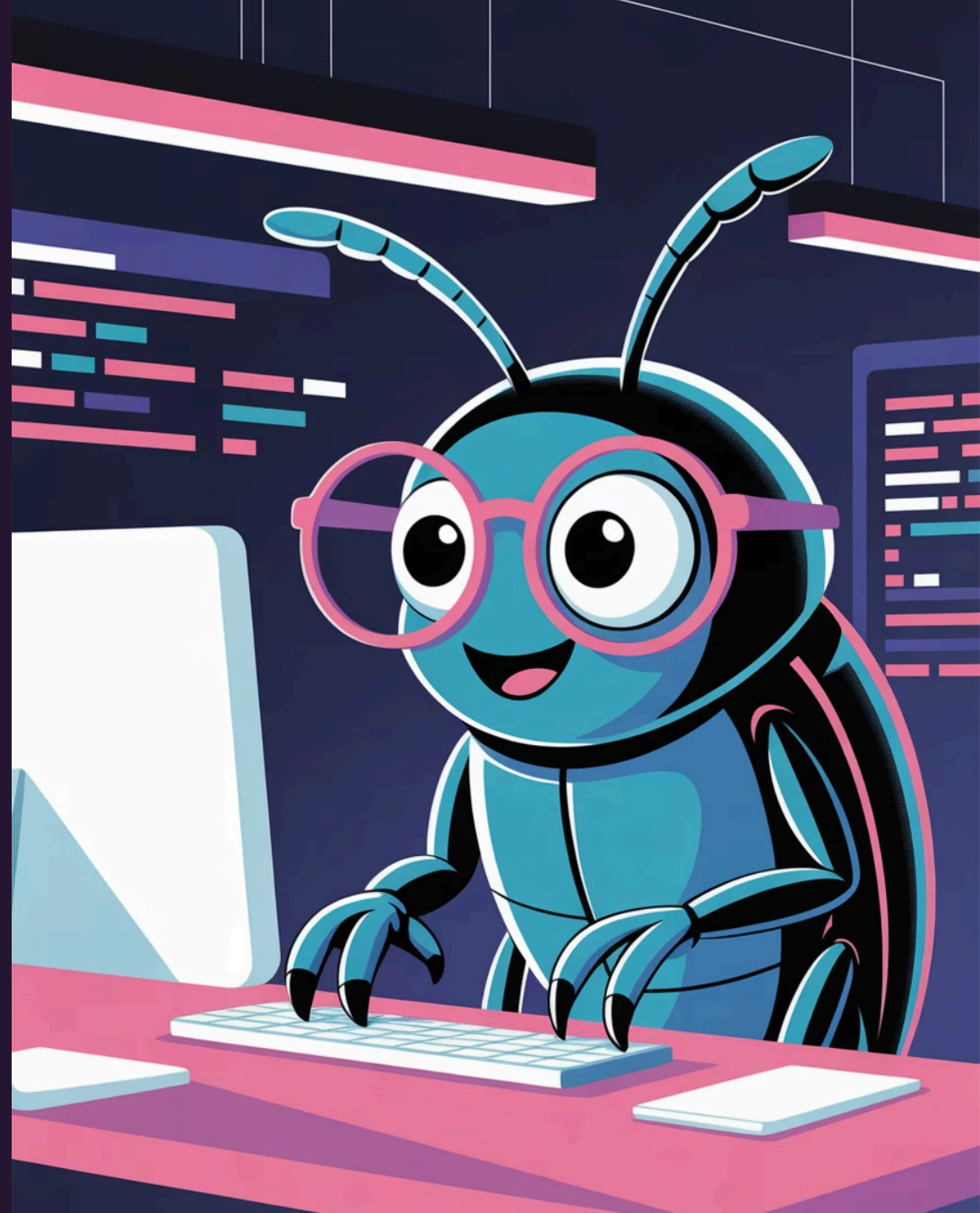


F.I.R.S.T Principles Overview

The *FIRST* acronym summarizes essential qualities of good unit tests

 by Rodrigo Martins Pagliares



The F.I.R.S.T Principles Overview

F = Fast

tests run quickly because they avoid networks and databases

I = Independent

each test stands alone and does not rely on others

R = Repeatable

running the same test multiple times must yield identical results

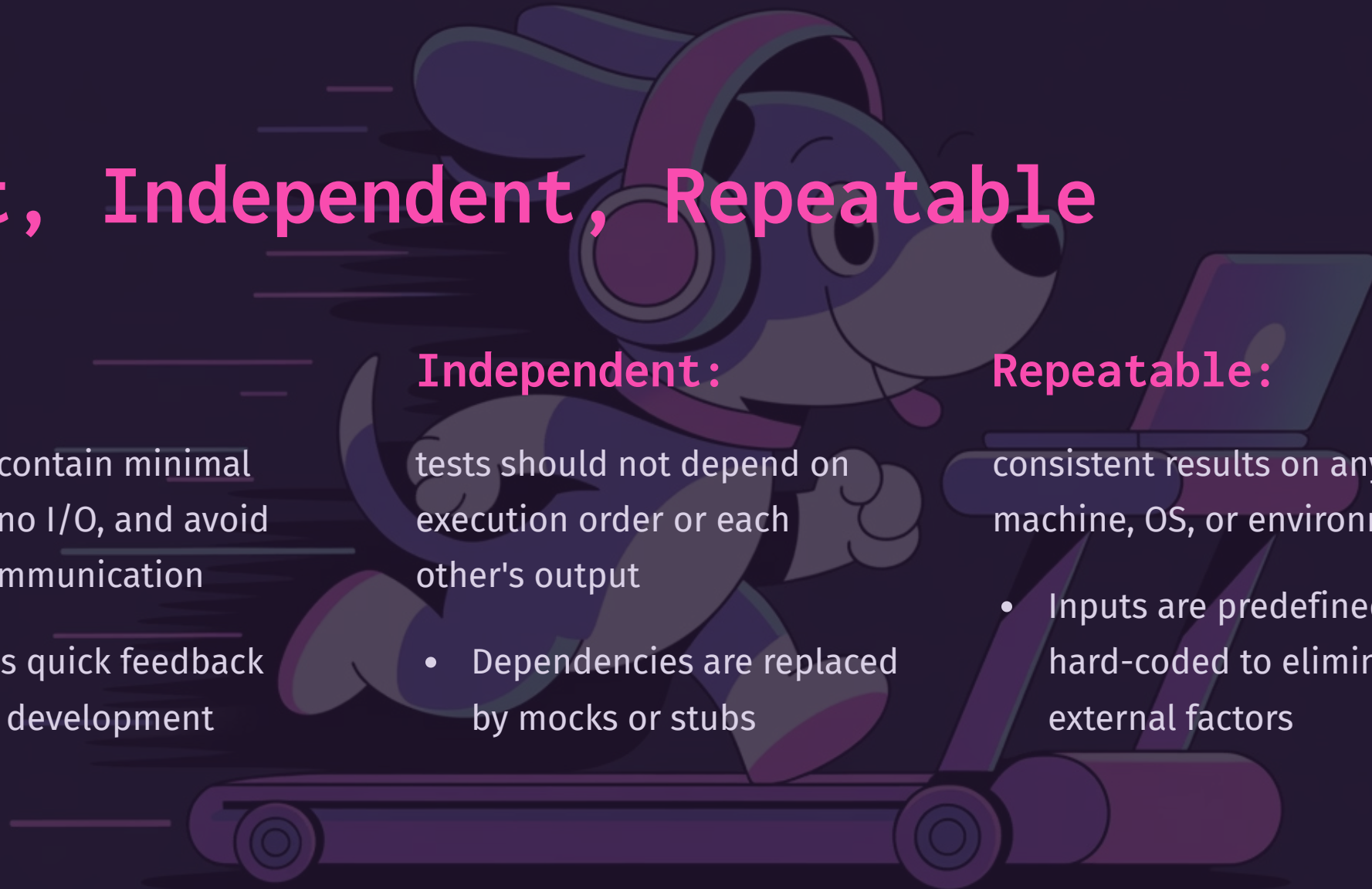
S = Self-Validating

tests automatically determine pass/fail states

T = Thorough

(or **Timely** in TDD contexts): tests cover happy and unhappy paths

Fast, Independent, Repeatable



Fast:

unit tests contain minimal logic, use no I/O, and avoid remote communication

- Ensures quick feedback during development

Independent:

tests should not depend on execution order or each other's output

- Dependencies are replaced by mocks or stubs

Repeatable:

consistent results on any machine, OS, or environment

- Inputs are predefined and hard-coded to eliminate external factors

Self-Validating and Thorough/Timely



Self-Validating:

no manual checks required
—**assertions** decide the outcome

- Results are clearly marked as pass (**green**) or fail (**red**)

Thorough:

multiple tests are created to cover:

- Valid inputs
- Invalid inputs
- Boundary values (min/max ranges)

Timely:

in TDD, tests are written as features are implemented

- Catch issues early and prevent bugs from reaching production



Conclusion

- F.I.R.S.T encapsulates the qualities of reliable, maintainable, and high-quality unit tests
 - Fast, independent tests improve productivity
 - Repeatability and self-validation ensure consistent, automated results
 - Thorough and timely testing prevents regressions and strengthens design
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Open Floor Discussion

- Which F.I.R.S.T principle do you currently struggle with the most?
- Have you tried writing timely tests following TDD?
- What strategies do you use to ensure independence between your unit tests?